

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI**Work Integrated Learning Programmes Division**

Cluster Programme - M. Tech in AI & ML

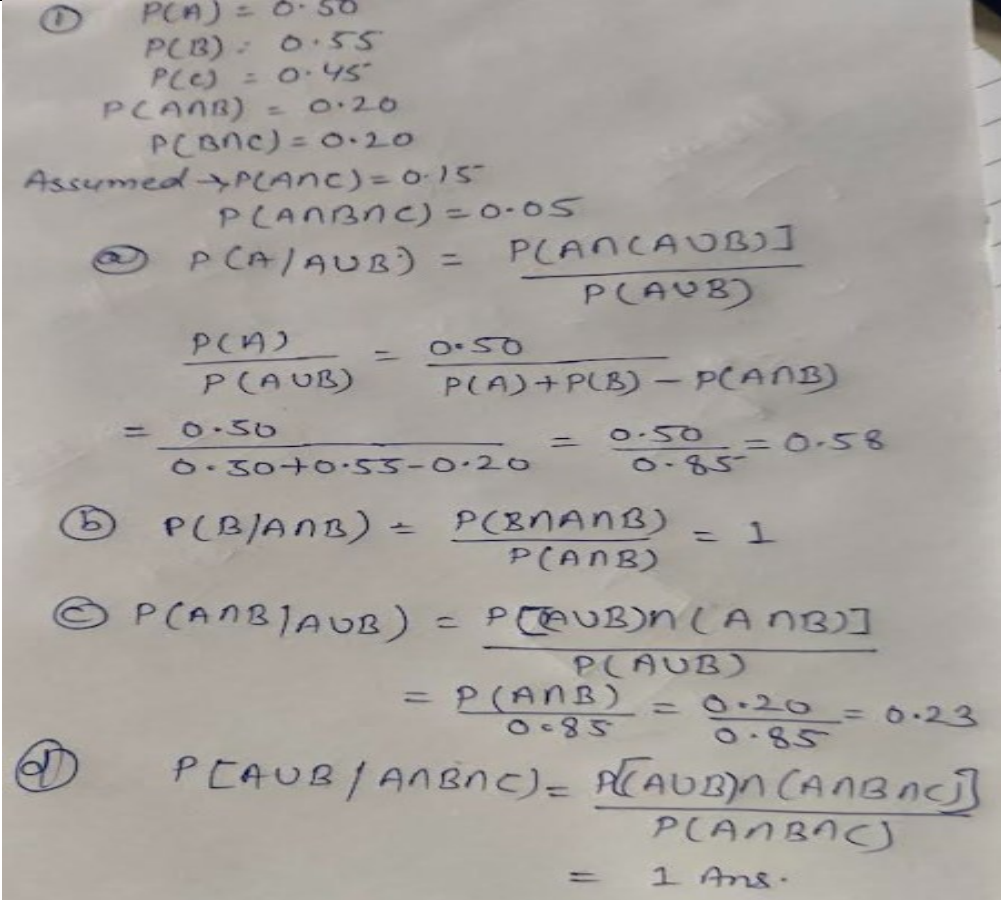
II Semester , 2023 – 24(July,2024)

Mid semester Examination (**Regular**)

Course No : AIMLC ZC418
 Course Title : Introduction to Statistical Methods
 Nature of Exam. : Open Book (Online)
 Weightage : 30 Marks
 Duration : 120 minutes
 Date : 20th July,2024_10 AM

*Number of questions:8**Number of Pages: 2***ANSWER KEY**

Q. No	Question	Marks																																													
Q.1.	Following is the statistical summary of some data set <table><tr><th></th><th>HHV</th><th>WBN</th><th>BNC</th><th>HBCN</th></tr><tr><td>Count</td><td>908</td><td>900</td><td>867</td><td>908</td></tr><tr><td>Mean</td><td>150</td><td>65</td><td>150</td><td>68</td></tr><tr><td>S.D</td><td>10</td><td>5</td><td>8</td><td>4</td></tr><tr><td>25%</td><td>90</td><td>25</td><td>90</td><td>30</td></tr><tr><td>50%</td><td>120</td><td>65</td><td>125</td><td>60</td></tr><tr><td>75%</td><td>130</td><td>70</td><td>135</td><td>63</td></tr><tr><td>Minimum</td><td>30</td><td>15</td><td>75</td><td>30</td></tr><tr><td>Maximum</td><td>160</td><td>75</td><td>180</td><td>90</td></tr></table> Write at least three inferences based on the above summary which helps in understanding the data		HHV	WBN	BNC	HBCN	Count	908	900	867	908	Mean	150	65	150	68	S.D	10	5	8	4	25%	90	25	90	30	50%	120	65	125	60	75%	130	70	135	63	Minimum	30	15	75	30	Maximum	160	75	180	90	3M
	HHV	WBN	BNC	HBCN																																											
Count	908	900	867	908																																											
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50%	120	65	125	60																																											
75%	130	70	135	63																																											
Minimum	30	15	75	30																																											
Maximum	160	75	180	90																																											
Q.1. Ans	1. To check consistency using coefficient of variation (CV= (sd/mean)x100): CV: HHV=6.6%, WBN=7.6%, BNC=5.3%, HBCN=5.8% Therefore, BNC data set is relatively more consistent with as low as only 5.3% of variation in the data around its mean. 2. To check symmetry(Mean=Median (50%)), positive skew (Mean>Median), negative skew (Mean<Median): HHV=Positive skew, WBN=Symmetric, BNC=Positive skew, HBCN=Positive skew Therefore, in WBN data set, most of the frequencies concentrate around the central part of the data whereas in the remaining data sets, they concentrate around the beginning (left extreme part of the series) part of the data. 3. To check whether minimum or/and maximum as potential outliers if they fall outside the range [Q1(25%)-1.5*(Q3-Q1), Q3(75%)+1.5*(Q3-Q1)]: HHV=[30, 190], WBN=[-50, 145], BNC=[22.5, 202.5], HBCN=[-19.5, 112.5] In HHV data set, its minimum term 30 might become a potential outlier which exactly fell on the left side border of the above limits whereas for the remaining data sets, their minimum and maximum terms would not become outliers.	(1 mark) (1 mark) (1 mark)																																													
Q.2	Let A and B are two independent events with probabilities 0.35 and 0.30 respectively. Then validate the following statements and justify a) $P(A \cup B) = 0$ as they are independent b) $P(A \cap B) = 0$ as they are mutually exclusive c) Find $P(A^c \cap B^c)$ (where A^c indicates the compliment of event A)	4M																																													

Q.2. Ans	a) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $P(A \cap B) = P(A) \cdot P(B) = 0.35 \times 0.3 = 0.105$ Thus $P(A \cup B) = 0.35 + 0.3 - 0.105 = 0.545$ (0.5Mark) So, the statement $P(A \cup B) = 0$ is incorrect. (0.5 Mark) b) Two events A and B are mutually exclusive if they cannot occur simultaneously, i.e., $P(A \cap B) = 0$. But $P(A \cap B) = P(A) \cdot P(B) = 0.35 \times 0.3 = 0.105$ (0.5 Mark) Therefore, the statement $P(A \cap B) = 0$ is incorrect (0.5 Mark) c) $P(A^c) = 1 - P(A) = 0.65$ and $P(B^c) = 1 - P(B) = 0.7$ (1 Mark) For independent events A and B, their complements A^c and B^c are also independent. $P(A^c \cap B^c) = P(A^c) \cdot P(B^c) = 0.65 \times 0.7 = 0.455$ (1Mark)	
Q.3	Consider the following probabilities of three events A, B and C: $P(A) = 0.50, P(B) = 0.55, P(C) = 0.45, P(A \cap B) = 0.20, P(B \cap C) = 0.20, P(A \cap C) = 0.15$ and $P(A \cap B \cap C) = 0.05$ Find the following a) $P(A A \cup B)$ b) $P(B A \cap B)$ c) $P(A \cap B A \cup B)$ d) $P(A \cup B A \cap B \cap C)$	4M
Q.3. Ans	 ① $P(A) = 0.50$ $P(B) = 0.55$ $P(C) = 0.45$ $P(A \cap B) = 0.20$ $P(B \cap C) = 0.20$ Assumed $\rightarrow P(A \cap C) = 0.15$ $P(A \cap B \cap C) = 0.05$ ② $P(A A \cup B) = \frac{P(A \cap (A \cup B))}{P(A \cup B)}$ $\frac{P(A)}{P(A \cup B)} = \frac{0.50}{P(A) + P(B) - P(A \cap B)}$ $= \frac{0.50}{0.50 + 0.55 - 0.20} = \frac{0.50}{0.85} = 0.58$ ③ $P(B A \cap B) = \frac{P(B \cap (A \cap B))}{P(A \cap B)} = 1$ ④ $P(A \cap B A \cup B) = \frac{P((A \cup B) \cap (A \cap B))}{P(A \cup B)}$ $= \frac{P(A \cap B)}{0.85} = \frac{0.20}{0.85} = 0.23$ ⑤ $P(A \cup B A \cap B \cap C) = \frac{P((A \cup B) \cap (A \cap B \cap C))}{P(A \cap B \cap C)}$ $= 1 \text{ Ans.}$	(1 mark) (1 mark) (1 mark) (1 mark)
Q.4	Hardik Pandya, Rishabh Pant and Surya Kumar Yadav are in the race of leading Indian cricket team in the next world cup with probabilities 0.2, 0.5 and 0.3 respectively. The probabilities of getting an increase in the match fee by	4M

	Hardik, Rishabh and Surya are 0.3, 0.6 and 0.5 respectively if they become the Captain. If there is an increase in match fee then find the probability - that it is because of Hardhik - that it is because of Rishabh - that it is because of Surya Kumar	
Q.4. Ans	Let's define the events: H: Hardik Pandya becomes the captain. R: Rishabh Pant becomes the captain. S: Surya Kumar Yadav becomes the captain. F: There is an increase in the match fee. Given probabilities are: $P(H)=0.2$, $P(R)=0.5$, $P(S)=0.3$ The conditional probabilities of getting an increase in the match fee given each player becomes the captain are: $P(F H) = 0.3$, $P(F R) = 0.6$, $P(F S) = 0.5$ By Law of total Probability $P(F) = P(H)P\left(\frac{F}{H}\right) + P(R)P\left(\frac{F}{R}\right) + P(S)P\left(\frac{F}{S}\right)$ $= 0.2 \times 0.3 + 0.5 \times 0.6 + 0.3 \times 0.5 = 0.17$ we use Bayes' theorem to find the probability that the increase in match fee is due to each specific player becoming the captain: (a) $P(H/F) = \frac{P(H)P(F/H)}{P(H)P(F/H)+P(R)P(F/R)+P(S)P(F/S)} = \frac{0.2 \times 0.3}{0.2 \times 0.3 + 0.5 \times 0.6 + 0.3 \times 0.5} = \frac{2}{17} = 0.1176$ (b) $P(R/F) = \frac{P(R)P(F/R)}{P(H)P(F/H)+P(R)P(F/R)+P(S)P(F/S)} = \frac{0.5 \times 0.6}{0.2 \times 0.3 + 0.5 \times 0.6 + 0.3 \times 0.5} = \frac{10}{17} = 0.5882$ (c) $P(S/F) = \frac{P(S)P(F/S)}{P(H)P(F/H)+P(R)P(F/R)+P(S)P(F/S)} = \frac{0.3 \times 0.5}{0.2 \times 0.3 + 0.5 \times 0.6 + 0.3 \times 0.5} = \frac{5}{17} = 0.2941$	(1 mark) (0.5 mark) (1mark) (0.5 mark) (0.5 mark) (0.5 mark)
Q.5	Let $f(x)$ be a continuous random variable defined in $[0,2]$ as $f(x) = k(x + 1)/4$, then find - k value - mean of x - Expectation of x^2 - Variance $P(0.5 < x < 1.5)$	4M

Q.5. Ans	a) $\int_0^2 \frac{k(x+1)}{4} dx = 1 \Rightarrow \frac{k}{4} \left[\frac{(x+1)^2}{2} \right]_0^2 = 1 \quad K=1$ b) $E(x) = \int_0^2 \frac{x(x+1)}{4} dx = \frac{1}{4} \left[\frac{x^3}{3} + \frac{x^2}{2} \right]_0^2 = \frac{7}{6} = 1.1666$ c) $E(x^2) = \int_0^2 \frac{x^2(x+1)}{4} dx = \frac{1}{4} \left[\frac{x^4}{4} + \frac{x^3}{3} \right]_0^2 = \frac{5}{3} = 1.6666$ d) $Var(x) = E(x^2) - [E(x)]^2 = \frac{11}{36} = 0.3055$ e) $P(0.5 < x < 1.5) = \int_{0.5}^{1.5} \frac{x+1}{4} dx$ $= \left[\frac{(x+1)^2}{8} \right]_{0.5}^{1.5} = 0.5$	0.5M 1M 1M 0.5M 1M																									
Q.6	Let X and Y are two independent random variables with the probability distributions given as <table><tr><td>X</td><td>-1</td><td>0</td><td>1</td><td>2</td></tr><tr><td>P(X)</td><td>0.25</td><td>0.15</td><td>0.35</td><td>0.25</td></tr></table> <table><tr><td>Y</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>P(Y)</td><td>0.10</td><td>0.20</td><td>0.30</td><td>0.40</td></tr></table> Then find the following a) If possible find the joint distribution of X and Y. If not, justify it. b) If possible ,find $P(x < 1,Y < 2)$ c) If possible, find $P(X < 1 / Y < 2)$	X	-1	0	1	2	P(X)	0.25	0.15	0.35	0.25	Y	0	1	2	3	P(Y)	0.10	0.20	0.30	0.40	4M					
X	-1	0	1	2																							
P(X)	0.25	0.15	0.35	0.25																							
Y	0	1	2	3																							
P(Y)	0.10	0.20	0.30	0.40																							
Q.6. Ans	Part (a): Joint Distribution of X and Y <table><tr><td>$P(X, Y)$</td><td>Y = 0</td><td>Y = 1</td><td>Y = 2</td><td>Y = 3</td></tr><tr><td>X = -1</td><td>0.25×0.10 = 0.025</td><td>0.25×0.20 = 0.05</td><td>0.25×0.30 = 0.075</td><td>0.25×0.40 = 0.10</td></tr><tr><td>X = 0</td><td>0.15×0.10 = 0.015</td><td>0.15×0.20 = 0.03</td><td>0.15×0.30 = 0.045</td><td>0.15×0.40 = 0.06</td></tr><tr><td>X = 1</td><td>0.35×0.10 = 0.035</td><td>0.35×0.20 = 0.07</td><td>0.35×0.30 = 0.105</td><td>0.35×0.40 = 0.14</td></tr><tr><td>X = 2</td><td>0.25×0.10 = 0.025</td><td>0.25×0.20 = 0.05</td><td>0.25×0.30 = 0.075</td><td>0.25×0.40 = 0.10</td></tr></table> Part (b): $P(X<1, Y<2)$ $P(X < 1,Y < 2) = P(X = -1,Y = 0) + P(X = -1,Y = 1) + P(X = 0,Y = 0) + P(X = 0,Y = 1)$ $= 0.025 + 0.05 + 0.015 + 0.03$ $= 0.12$	$P(X, Y)$	Y = 0	Y = 1	Y = 2	Y = 3	X = -1	0.25×0.10 = 0.025	0.25×0.20 = 0.05	0.25×0.30 = 0.075	0.25×0.40 = 0.10	X = 0	0.15×0.10 = 0.015	0.15×0.20 = 0.03	0.15×0.30 = 0.045	0.15×0.40 = 0.06	X = 1	0.35×0.10 = 0.035	0.35×0.20 = 0.07	0.35×0.30 = 0.105	0.35×0.40 = 0.14	X = 2	0.25×0.10 = 0.025	0.25×0.20 = 0.05	0.25×0.30 = 0.075	0.25×0.40 = 0.10	(2Mark) (1Mark) (1Mark)
$P(X, Y)$	Y = 0	Y = 1	Y = 2	Y = 3																							
X = -1	0.25×0.10 = 0.025	0.25×0.20 = 0.05	0.25×0.30 = 0.075	0.25×0.40 = 0.10																							
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X = 2	0.25×0.10 = 0.025	0.25×0.20 = 0.05	0.25×0.30 = 0.075	0.25×0.40 = 0.10																							

	C) $P(X < 1 Y < 2)$ $P(X < 1 Y < 2) = \frac{P(X < 1 \text{ and } Y < 2)}{P(Y < 2)}$ We already calculated $P(X < 1 \text{ and } Y < 2) = 0.12$. Next, we need to find $P(Y < 2)$: $P(Y < 2) = P(Y = 0) + P(Y = 1) = 0.10 + 0.20 = 0.30$ Now, we can calculate the conditional probability: $\begin{aligned} P(X < 1 Y < 2) &= \frac{P(X < 1 \text{ and } Y < 2)}{P(Y < 2)} \\ &= \frac{0.12}{0.30} \\ &= 0.4 \end{aligned}$	
Q.7	Let X be a random variable which follows binomial distribution with $n = 150$ and $p = 0.2$ then find $P(50 < X < 80)$	4M
Q.7. Ans	Solution: To find $P(50 < X < 80)$ where X is a binomial random variable with parameters $n = 150$ and $p = 0.2$, we can use the normal approximation to the binomial distribution because n is large. The mean and standard deviation of X are : $\mu = np = 150 \times 0.2 = 30 \text{ and } \sigma = \sqrt{npq} = \sqrt{150 \times 0.2 \times 0.8} = \sqrt{24} = 4.89$ _____ (1 Mark) Using continuity correction, we can write $P(50 < X < 80) = P(50.5 < X < 79.5)$ Therefore, $\text{Std unit of } 50.5 = \frac{50.5 - 30}{4.89} = 4.19 \text{ and std unit of } 79.5 = \frac{79.5 - 30}{4.89} = 10.12$ _____ (1 Mark) So, $P(50 < X < 80) = P(50.5 < X < 79.5) = P(4.19 < Z < 10.12) = \phi(10.12) - \phi(4.19)$ But by normal distribution table $\phi(10.12) = \phi(4.19) \approx 1$. Hence $P(50 < X < 80) = P(50.5 < X < 79.5) = P(4.19 < Z < 10.12) = \phi(10.12) - \phi(4.19) = 1 - 1 = 0$ _____ (2 Marks) Note: If any students not considering continuity factor then deduct 1 mark only.	
Q.8	It is observed that average amount of time that a customer spends in a super market is 40 minutes with a standard deviation of 5 minutes. Then find the probability of sampling distribution of mean time lies between 35 and 45 minutes where the sample size is 50.	3M
Q.8. Ans	$\begin{aligned} &P(35 < \bar{X} < 45) \\ &= P\left(\frac{35 - 40}{5/\sqrt{50}} < Z < \frac{45 - 40}{5/\sqrt{50}}\right) \\ &= P(-7.07 < Z < 7.07) \\ &= P(Z < 7.07) - P(Z < -7.07) \\ &\approx 1 - 0 \end{aligned}$	(1 MARK) (1 MARK)

	≈ 1 Note: Referring to the Z table values which ranges from -4 to 3. Z beyond -4 approaches 0 and beyond 3 is approaches 1.	(1 MARK)
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